

Introduction to the Laws of Motion

Lesson 6: Newton's Third Law

Newton's Third Law - Study Notes

Key Concepts

1. **Action–Reaction Pairs** – Every force exerted by one object on another is met with an equal and opposite force.
2. **Equal Magnitude, Opposite Direction** – The two forces have the same size (newtons) but act in opposite directions.
3. **Different Objects** – The forces act on *different* bodies; they never cancel on the same object.
4. **Simultaneity** – The action and reaction occur at the same instant.
5. **Conservation of Momentum** – Action–reaction pairs are the reason total momentum of an isolated system is conserved.

Summary

Newton's Third Law states that for every force there is a **reaction force** that is equal in magnitude and opposite in direction. The law emphasizes the interaction between two objects: if object /A pushes on object /B, object /B pushes back on object /A with the same strength. Because the forces act on different objects, they do not cancel each other out; each object experiences its own net force, which determines its motion according to Newton's Second Law.

Understanding this principle helps explain everyday phenomena—from why we can walk without slipping to how rockets launch into space. In each case, a body exerts a force on its surroundings, and the surroundings exert an equal and opposite force back on the body, producing the observed motion.

Important Points

- **Force Pairs:** Write them as " $\vec{F}_{AB}$ on B = $-\vec{F}_{BA}$ on A".
- **No Self Cancellation:** Forces in a pair act on **different** objects, so they never cancel on a single object.
- **Mass Matters:** The resulting accelerations differ because $a = F/m$; the lighter object accelerates more.
- **Internal vs. External:** In a closed system, internal action–reaction forces cancel, leaving only external forces to change the system's momentum.
- **Common Misconception:** The "reaction" is not a delayed response; it is simultaneous with the "action".

Examples

1. Rocket Launch

- **Action:** Hot gases are expelled downward through the rocket nozzle (force on the gases).

- **Reaction:** The gases push upward on the rocket with an equal and opposite force, propelling it skyward.

- **Explanation:** The rocket's mass is much larger than the expelled gases, so the rocket's upward acceleration is smaller but sufficient to overcome gravity.

2. Walking

- **Action:** Your foot pushes backward against the ground.

- **Reaction:** The ground pushes forward on your foot with an equal and opposite force, moving you ahead.

- **Explanation:** Because the ground is effectively immovable (very large mass), you experience a noticeable forward acceleration while the ground's motion is negligible.

Quick Review

1. **What does Newton's Third Law say about the direction of the reaction force?**
2. **Why don't action–reaction forces cancel each other out on a single object?**
3. **In a rocket, which object experiences the larger acceleration: the rocket or the expelled gases? Why?**
4. **How does walking illustrate the concept of an action–reaction pair?**
5. **If two ice skaters push off each other, what determines how fast each skater moves away?**

Further Reading

- Newton's First and Second Laws (foundation for dynamics)
- Conservation of Momentum in Collisions
- Friction and Normal Forces (how they affect action–reaction pairs)
- Real world applications: jet engines, propellers, and sports biomechanics
- Vector representation of forces and free body diagrams.

